



ML Model Evaluation for AQI Prediction in IGP Using Air Quality and Meteorological Features

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Introduction

Artificial Intelligence (AI) enables systems to learn and decide autonomously (Xu et al., 2021), with Machine Learning (ML) identifying patterns without explicit programming (Liu, 2018). Feature engineering enhances model performance by converting raw data into meaningful inputs.

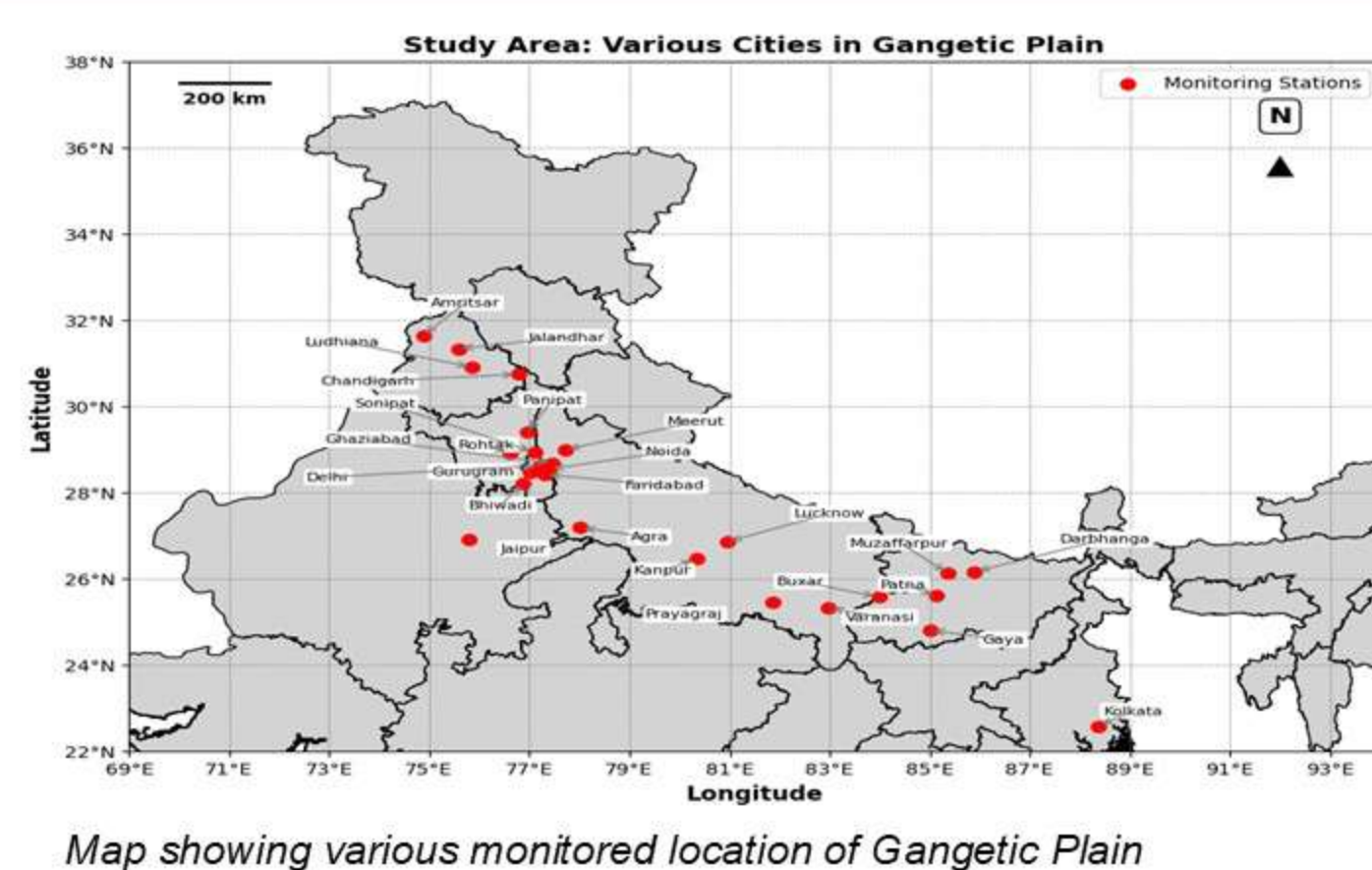
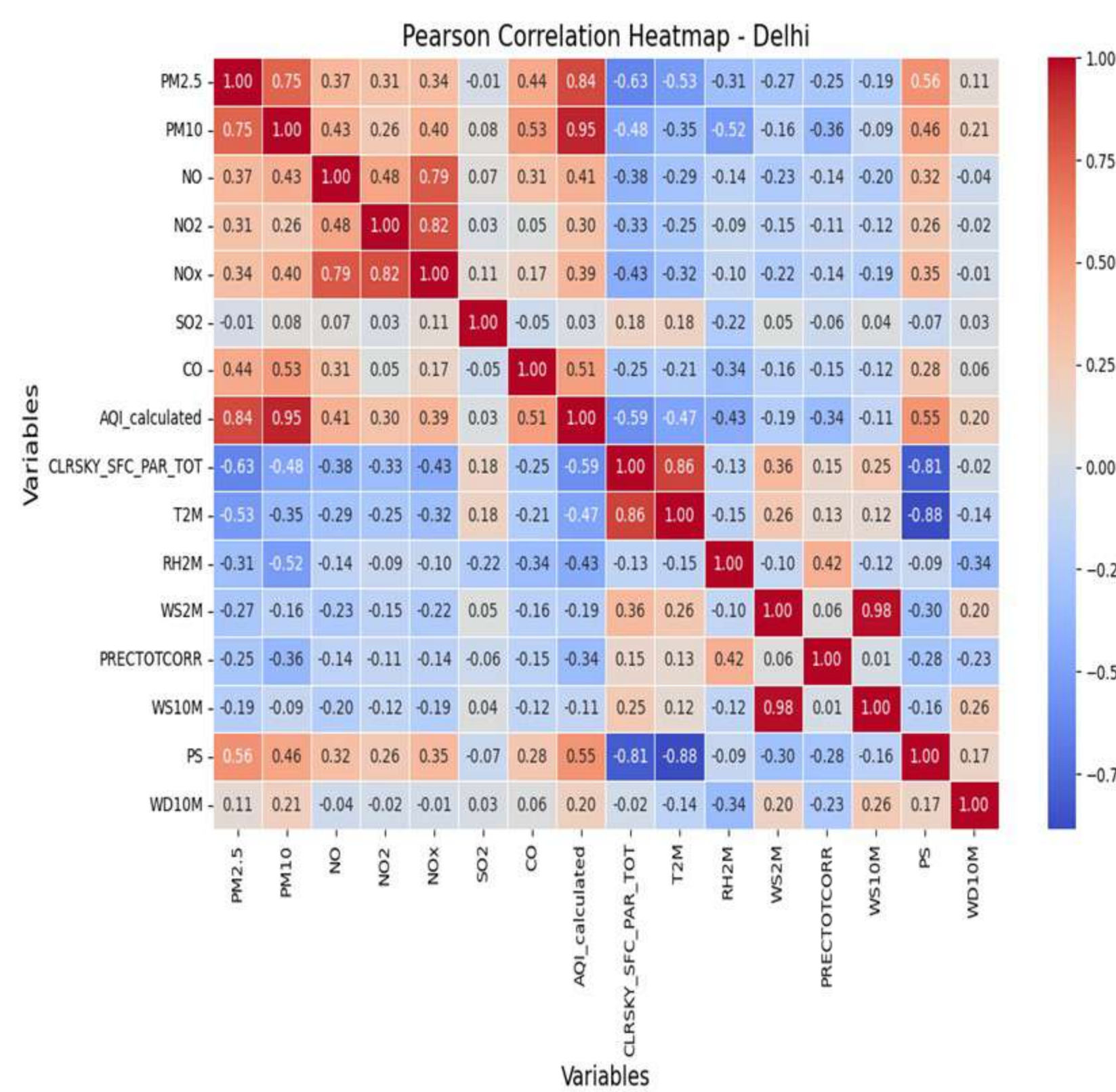
Air pollution causes 4.2–8.9 million deaths annually (Thakur et al., 2024), severely impacting regions like the Gangetic Plain (Chandra et al., 2022). Most AQI models focus on pollutants, but this study incorporates meteorological factors (temperature, humidity, wind, pressure) to improve accuracy (Mishra & Patel, 2024). Unlike traditional models based on chemical equations (Vlasenko et al., 2021), ML offers faster, scalable predictions (Xing et al., 2020). Ensemble techniques like Random Forest and stacking enhance generalization and reduce overfitting (Geurts et al., 2006; Zhang & Ma, 2012). Success in countries like China shows the importance of accurate AQI forecasting for policy (Zhang et al., 2019a). This study uses R^2 , MAE, RMSE, and MAPE metrics (Botchkarev, 2018) to support sustainable air quality management.

Methodology

Study area:

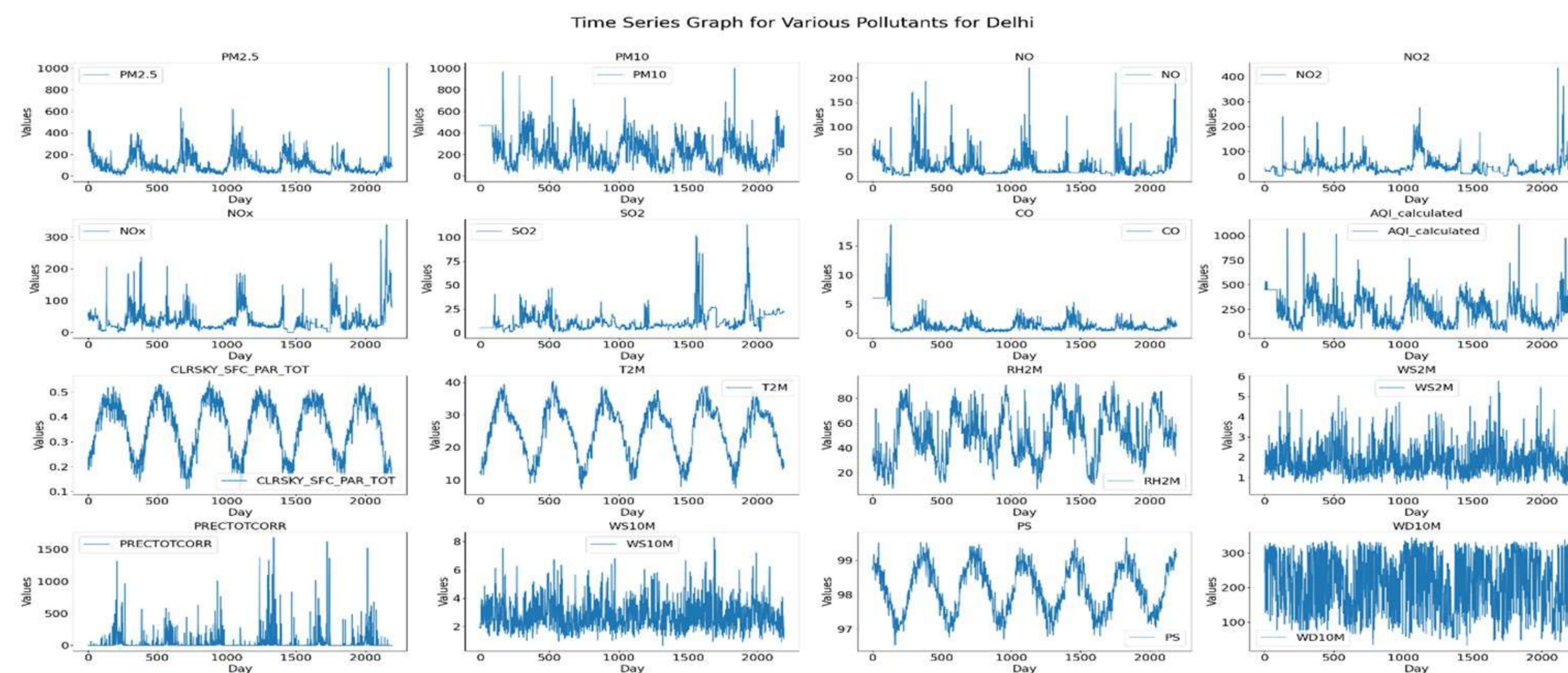
The study covers 25 cities across the Gangetic Plain.

Atmospheric & Meteorological Variables:



Map showing various monitored location of Gangetic Plain

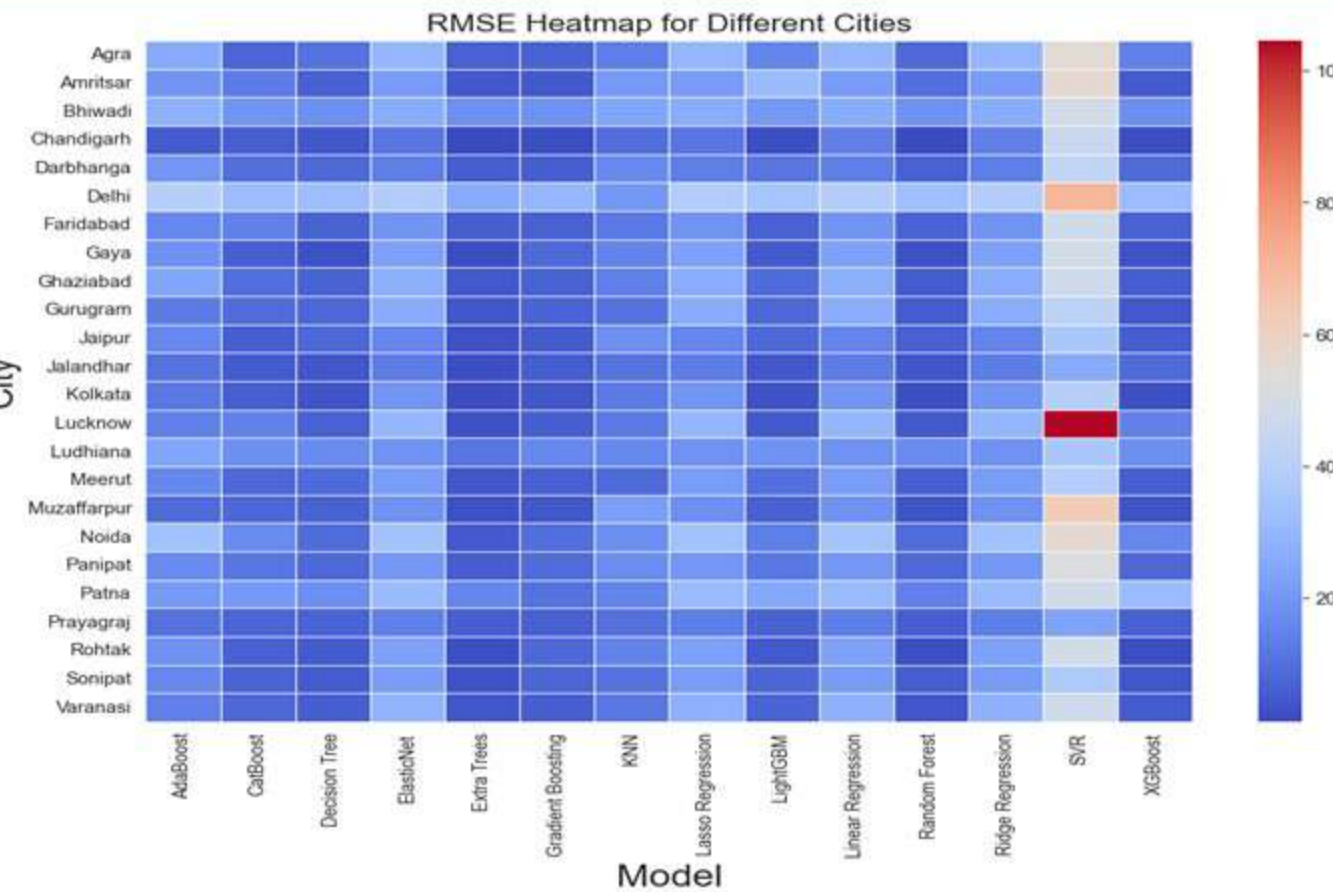
- Strong positive correlation between AQI & PM2.5 (0.90) and PM10 (0.85)
- Moderate positive correlation (50-55%) with CO and surface pressure- suggests pollutant trapping under high pressure.
- Strong negative correlation (60%) with CLRSKY_SFC_PAR_TOT solar radiation may aid pollutant dispersion via photochemical processes.
- Negative with temperature and relative humidity (RH2M)- higher temp & RH help disperse or remove pollutants.
- Negative correlation (35%) with precipitation- rainfall helps clean air by washing out pollutants.



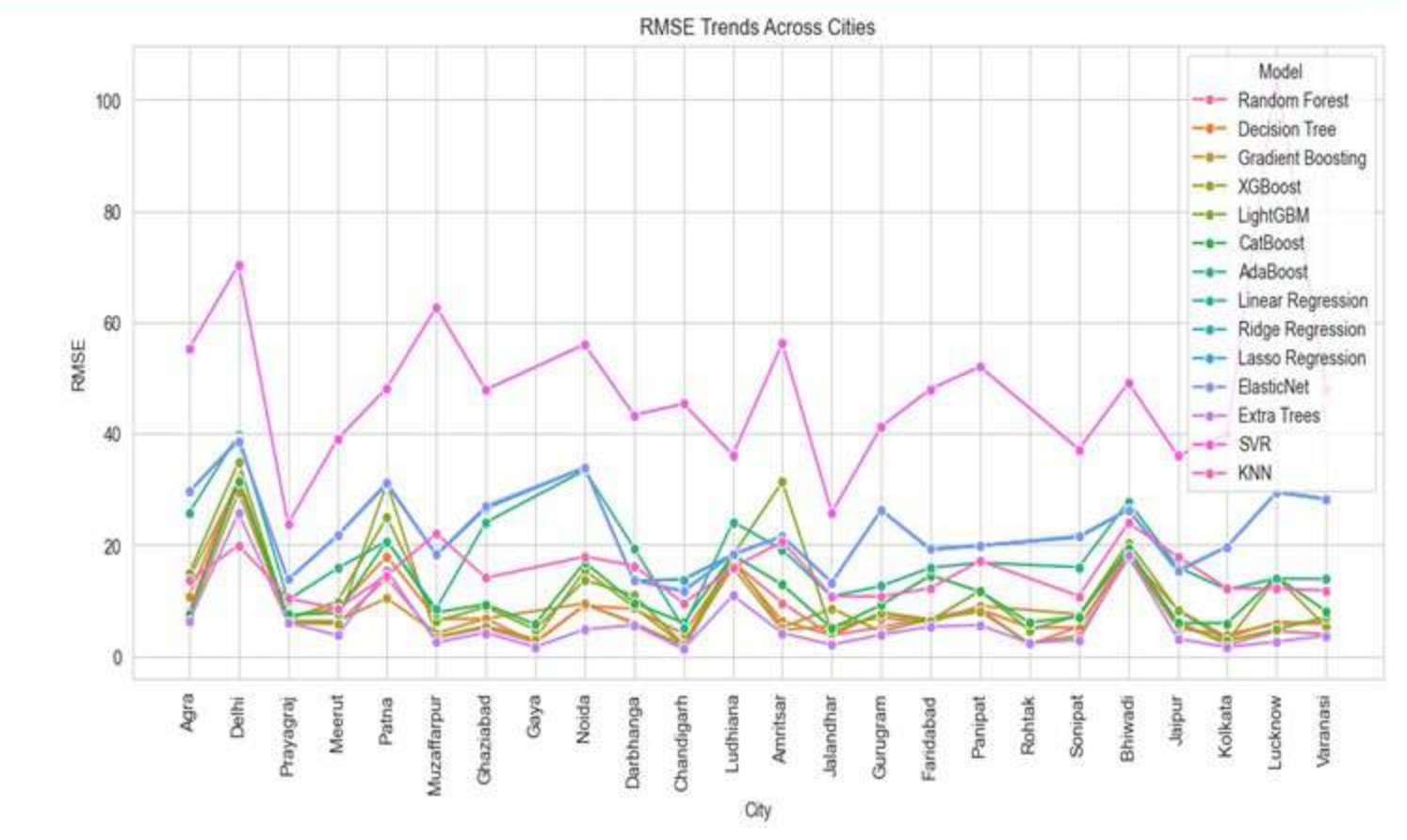
References

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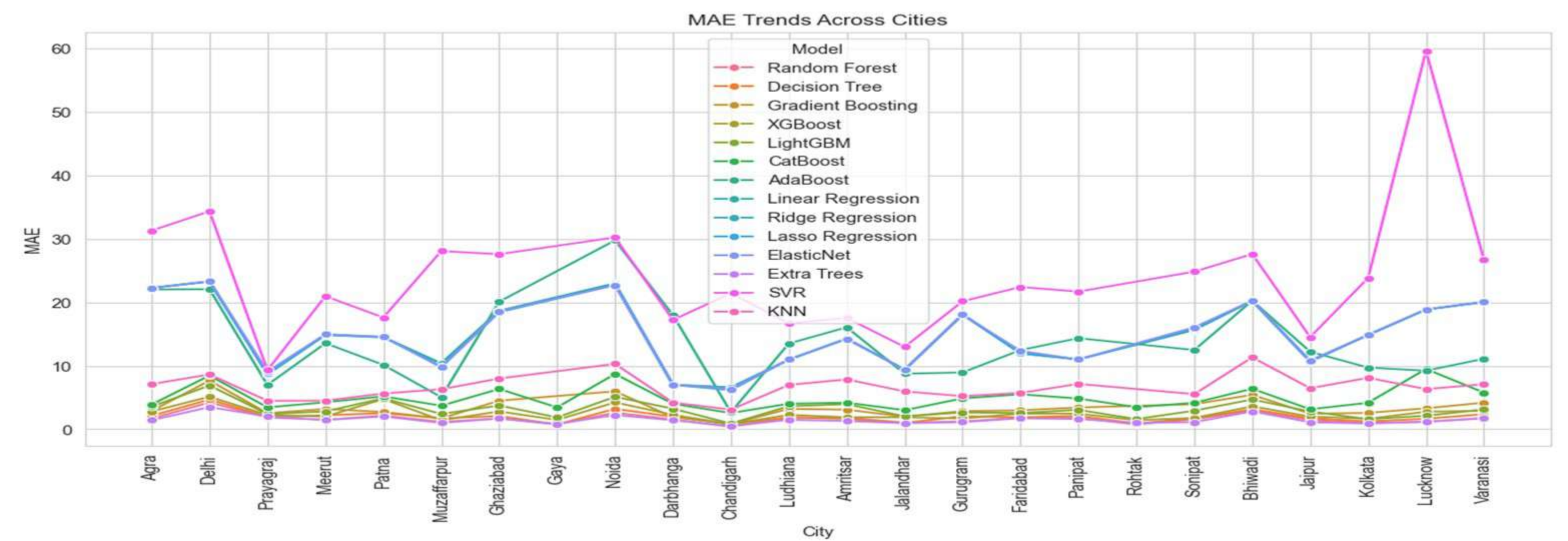
Results and Performance Evaluation



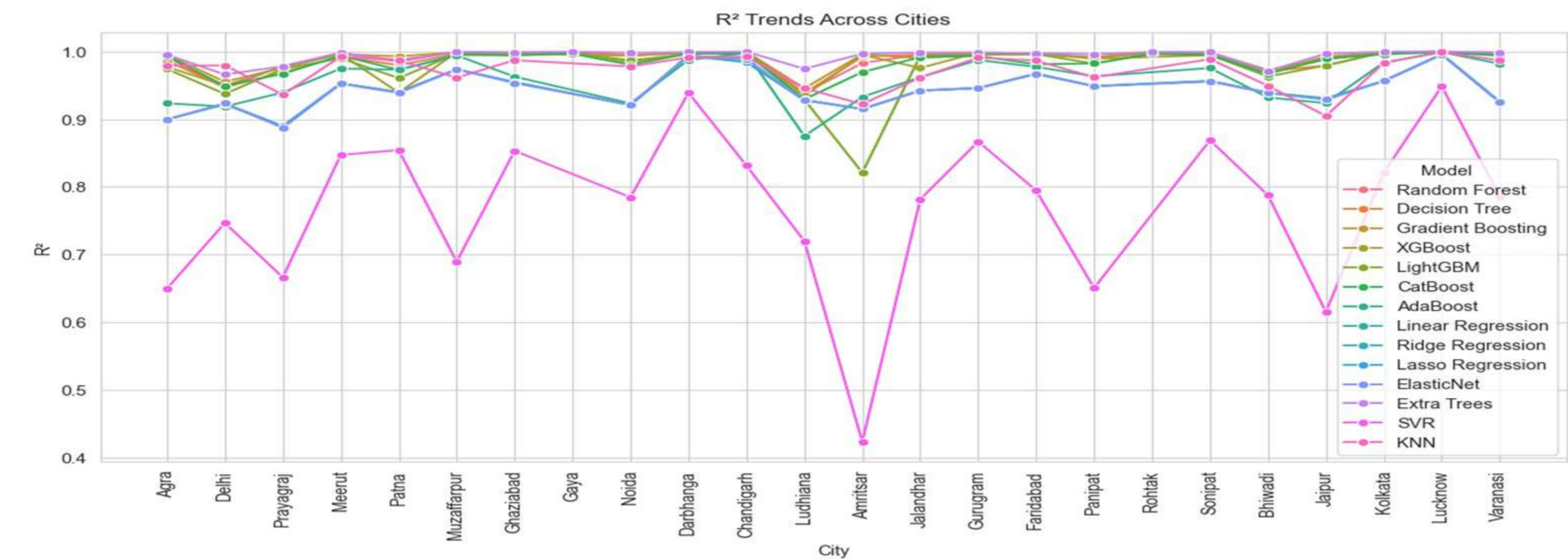
Heatmap showing RMSE values of different models with all the cities



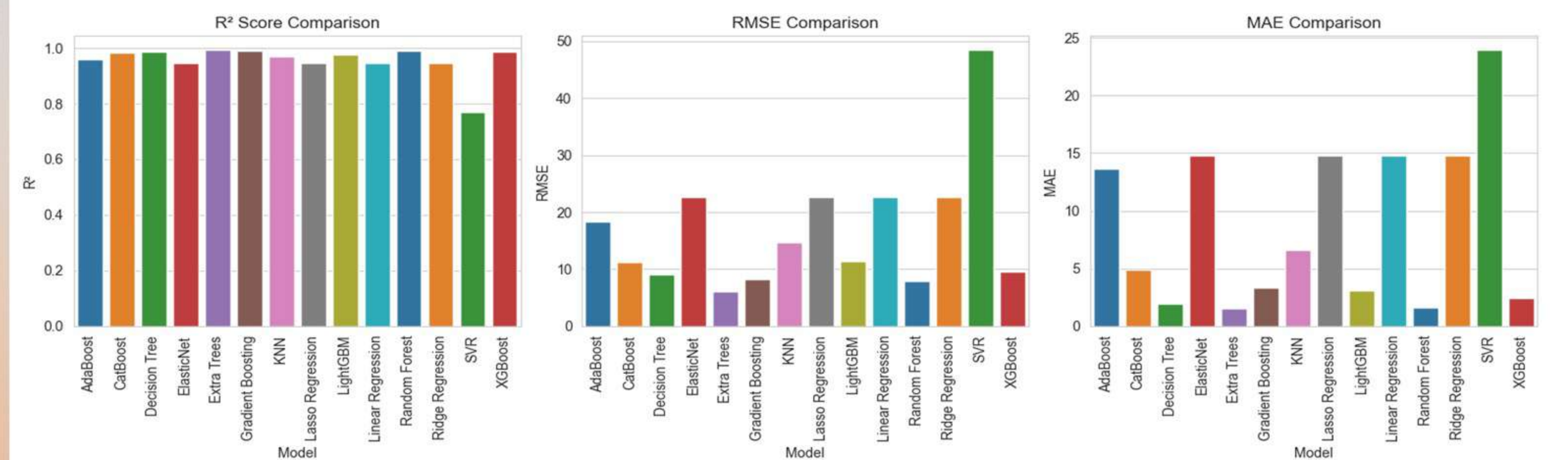
RMSE for all cities for all models



MAE values for all cities for all models



R² score for all cities for all models



Histograms showing performance metrics, averaged for all the cities

Conclusion

- Ensemble models (Extra Trees, Random Forest, XGBoost, LightGBM) achieved the best performance with low errors and high accuracy.
- Boosting models (CatBoost, Gradient Boosting) also showed strong predictive results.
- Linear models (Linear, Ridge, Lasso, ElasticNet) and SVR performed poorly, especially in complex cities like Lucknow and Varanasi.
- SVR was the weakest due to its inability to handle nonlinear relationships in air quality data.
- Air pollution data is highly dynamic and nonlinear, influenced by weather, geography, and variable interactions.
- Linear models struggle due to their assumption of linear dependencies, and are sensitive to multicollinearity, missing data, and outliers.
- Residual analysis confirmed that ensemble models provide stable and reliable predictions.
- High RMSE in cities like Muzaffarpur and Bhiwadi highlights challenges from environmental variability.
- Conclusion: Ensemble methods are the most effective for PM2.5 forecasting due to their robustness and superior generalization.